

$f(x) = \frac{(x^2-1)}{(x-1)}$ ອະ ກິດ ທີ່ ມີ ກິດ ພຽກ ມີ ຄ່າ ບໍ່ ມີ
 ອະ ກິດ ພຽກ ມີ ຄ່າ $x=1$ ຈຶ່ງ ບໍ່ ມີ ຄ່າ ບໍ່ ມີ

$$f_{02} = 0\%$$

$\frac{d^2y}{dx^2} = \frac{dy}{dx}$

$$f(x) = \frac{x^0}{x+1} \quad 0000, \quad 001, \quad 010, \quad 100, \quad 101$$

200000 2000000

$$f(x) = \frac{x^2}{x^2 + 1}$$

2. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ (1/4)

$\frac{\%}{\%} \cdot \frac{\%}{\%} \cdot \frac{\%}{\%} \dots \dots$
 සියලුම මග පෙන්වන
 මග, පැමිණිලි, අවස්ථා, සහ, මෙහි අවස්ථා වල
 ම අවස්ථා අතර අන්තර්ගත වන මග සියලුම
 මග සම්බන්ධයෙන් සෑදීමට හැකි, සියලුම මග සම්බන්ධයෙන්
 වැඩේ.

(ii) $f(x) = \frac{x^2 - 1}{x - 1}$ $\lim_{x \rightarrow 1} f(x) = \frac{0}{0}$ $x \rightarrow 1$

[illegible]

$0.9 \quad 0.99 \quad 0.999 \quad 0.9999$					$1.00001 \quad 1.0001 \quad 1.001 \quad 1.01 \quad 1.1$				
x	0.9	0.99	0.999	0.9999	x	1.1	1.01	1.001	1.0001
$f(x)$	1.9	1.99	1.999	1.9999	$f(x)$	2.1	2.01	2.001	2.0001

$$x \rightarrow 100, f(x) \rightarrow 200$$

$$x = 0.9$$

$$f(x) = \frac{x^2 - 1}{x - 1}$$

$$f(0.9) = \frac{(0.9)^2 - 1^2}{0.9 - 1}$$

$$f(0.9) = \frac{[(0.9) - 1][(0.9) + 1]}{(0.9 - 1)}$$

$$f(0.9) = 1.9$$

$\therefore x \rightarrow 1^-$ හි $x \rightarrow 2$

$$f(x) = \frac{x^2 - 1}{x - 1}$$

$$f(1.1) = \frac{(1.1)^2 - 1^2}{1.1 - 1}$$

$$f(1.1) = \frac{[(1.1) - 1][(1.1) + 1]}{(1.1 - 1)}$$

Q2) $f(x) = \frac{x^2 - 4}{x - 2}$ 20 අඩු අගයන් හා 20 වැඩි අගයන්
යන ලෙසින් නිවැරදි ගණනය කිරීම.

$x \rightarrow 2$ හි $f(x)$ හි නිවැරදි අගය.

1.9 1.99 1.999 1.9999 2 2.0001 2.001 2.01 2.1

x	1.9	1.99	1.999	1.9999	x	2.1	2.01	2.001	2.0001
$f(x)$	3.9	3.99	3.999	3.9999	$f(x)$	4.1	4.01	4.001	4.0001

$x \rightarrow 2^-$ හි, $f(x) \rightarrow 4$ හි. $x \rightarrow 2^+$ $f(x) \rightarrow 4$ හි.

$$x = 1.950$$

$$f(1.9) = \frac{x^2 - 4}{x - 2}$$

$$= \frac{(1.9)^2 - 4}{1.9 - 2}$$

$$= \frac{(1.9 - 2)(1.9 + 2)}{1.9 - 2}$$

$$= 3.9$$

$$x = 1.9950$$

$$f(1.99) = \frac{x^2 - 4}{x - 2}$$

$$= \frac{(1.99)^2 - 4}{1.99 - 2}$$

$$= \frac{(1.99 - 2)(1.99 + 2)}{(1.99 - 2)}$$

$$= 3.99$$

$$x = 2.050$$

$$f(2.1) = \frac{x^2 - 4}{x - 2}$$

$$= \frac{(2.1)^2 - 4}{2.1 - 2}$$

$$= \frac{(2.1 - 2)(2.1 + 2)}{(2.1 - 2)}$$

$$= 4.1$$

$$x = 2.0150$$

$$f(2.01) = \frac{x^2 - 4}{x - 2}$$

$$= \frac{(2.01)^2 - 4}{2.01 - 2}$$

$$= \frac{(2.01 - 2)(2.01 + 2)}{(2.01 - 2)}$$

$$= 4.01$$

$\therefore x \rightarrow 2$ හි, $f(x) \rightarrow 4$ හි.

Q1) $f(x) = \frac{x^2 - 4}{x - 2}$ පිරිසිදු කිරීම $x = 2$ දී අර්ථ නොදැක්වෙන
 බව පෙන්වන්න.

(i) x වටය පමණක් 2 ට ආසන්න වන විට $f(x)$ හි අගයය
 සොයන්න.

(ii) x දුරස්ත පමණක් 2 ට ආසන්න වන විට $f(x)$ හි අගයය
 සොයන්න.

(iii) පවතින $x \rightarrow 2$ විට $f(x)$ හි අගයය සොයන්න.

(i) $f(x) = \frac{x^2 - 4}{x - 2}$

$$\lim_{x \rightarrow 2} = \frac{2^2 - 4}{2 - 2}$$

$$= \frac{0}{0}$$

$$= \frac{0}{0}$$

(ii) $f(x) = \frac{x^2 - 4}{x}$



$\therefore f(x)$ $x = 2$ දී අර්ථ නොදැක්වෙන.

$$x \rightarrow 2^- \text{ විට, } f(x) \rightarrow 4$$

$$x \rightarrow 2^+ \text{ විට, } f(x) \rightarrow 4$$

$\therefore x \rightarrow 2$ විට, $f(x) \rightarrow 4$ වේ.

මෙම පිටුව පිරිසිදු කිරීම.

$$\lim_{x \rightarrow 2} f(x) = 4$$

$$\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x - 2} \right) = 4$$

Limit $f(x) = l$ as $x \rightarrow a^-$ and Limit $f(x) = l$ as $x \rightarrow a^+$

$$\lim_{x \rightarrow a} f(x) = l.$$
[illegible]

$$\begin{aligned} & \textcircled{Q2} \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{x^2 - 2^2}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} \\ &= \lim_{x \rightarrow 2} (x+2) \\ &= 4 \end{aligned}$$

$$\begin{array}{llll} \textcircled{1} \lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} & \textcircled{2} \lim_{x \rightarrow 1} \frac{x - 1}{x^3 - 1} & \textcircled{3} \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2} & \textcircled{4} \lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x^2 - 4x + 3} \\ = \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x^2+x+1)} & = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)(x+2)} & = \lim_{x \rightarrow 1} \frac{(x-1)(x-2)}{(x-3)(x-1)} \\ = \lim_{x \rightarrow 1} \frac{x^2+x+1}{x+1} & = \lim_{x \rightarrow 1} \frac{1}{(x^2+x+1)} & = \frac{4/3}{1/2} & = \frac{1/2}{1/2} \\ = \frac{3/2}{1} & = \frac{1}{3} & = \frac{4}{3} & = \frac{1}{2} \end{array}$$

$$(x-2)(x^2+4)$$

$$*5 \lim_{x \rightarrow 1} \frac{\sqrt{x}-1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}}-1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}}-1}{(x^{\frac{1}{2}})^2-1^2}$$

$$= \lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}}-1}{(x^{\frac{1}{2}}-1)(x^{\frac{1}{2}}+1)} = \frac{1}{2}$$

$$*6 \lim_{x \rightarrow 1} \frac{3\sqrt{x}-1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{3x^{\frac{1}{2}}-1}{(x^{\frac{1}{2}})^3-1^3}$$

$$= \lim_{x \rightarrow 1} \frac{3x^{\frac{1}{2}}-1}{(x^{\frac{1}{2}}-1)(x^{\frac{1}{2}}+1)^2}$$

$$= \frac{3}{4}$$

$$*7 \lim_{x \rightarrow 2} \frac{x^3-8}{x^4-16}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x^2-2^2)(x^2+2^2)}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)(x^2+2^2)}$$

$$= \frac{4+4+4}{4(8)} = \frac{12}{32} = \frac{3}{8}$$

$$*8 \lim_{x \rightarrow 1} \frac{x^3+x^2+x-3}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{x^3+x^2+x-3}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{x^3-1+x^2+x-3+1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{x^3-1}{x-1} + \lim_{x \rightarrow 1} \frac{x^2+x-3+1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{x-1} + \lim_{x \rightarrow 1} \frac{(x-1)(x+2)}{x-1}$$

$$= \lim_{x \rightarrow 1} (x^2+x+1) + \lim_{x \rightarrow 1} (x+2)$$

$$= (1+1+1) + (1+2) = 3 + 3 = 6$$

$$= 6$$

$$9 \lim_{x \rightarrow -1} \frac{x^3+1}{x+1}$$

$$= \lim_{x \rightarrow -1} \frac{(x+1)(x^2-x+1)}{x+1}$$

$$= \lim_{x \rightarrow -1} (x^2-x+1)$$

$$= 1+1+1 = 3$$

$$10 \lim_{x \rightarrow 0} \frac{ax^2+bx+c}{cx^2+bx+a}$$

$$= \lim_{x \rightarrow 0} \frac{ax^2+bx+c}{cx^2+bx+a}$$

$$= \frac{c}{a}$$

$$7 \lim_{x \rightarrow 2} \frac{x^3-8}{x^4-16}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x^2-2^2)(x^2+2^2)}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)(x^2+2^2)}$$

$$= \lim_{x \rightarrow 2} \frac{x^2+2x+4}{(x+2)(x^2+2^2)}$$

$$= \frac{4+4+4}{4(8)} = \frac{12}{32} = \frac{3}{8}$$

$$= \frac{3}{8}$$

Q1) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

$f(x) = x^n - a^n$
 $x = a$ විට,
 $f(a) = a^n - a^n$
 $f(a) = 0$

$\therefore (x-a) f(x)$ ආකාරයේ වේ.

$$x-a \left[\begin{array}{c} x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1} \\ x^n - a^n \\ x^n - ax^{n-1} \end{array} \right]$$

$$= a^n + ax^{n-1} - ax^{n-1} - a^n$$

$$= a^{n-1}x - a^{n-1}x$$

$$\frac{x^n - a^n}{x - a} = x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1}$$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = \lim_{x \rightarrow a} (x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1})$$

↓
0

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = a^{n-1} + a \cdot a^{n-2} + a^2 \cdot a^{n-3} + \dots + a^{n-1}$$

$$= a^{n-1} + a^{n-1} + a^{n-1} + \dots + a^{n-1}$$

$$= na^{n-1}$$

$\therefore \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$

මෙම n ඉතා ඉහළ (∞)

* $\frac{x^n - a^n}{x - a} = na^{n-1}$

$n (-)$ ଲିମିଟର ଓ ଉକ୍ତ ପ୍ରକ୍ରିୟା ଓ ଫଳାଫଳ

$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$ ଯେଉଁ $n \in \mathbb{Z}^+$

$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$ ପ୍ରକାଶନ ଉପାଦାନ

$$\lim_{x \rightarrow a} \frac{x^{-n} - a^{-n}}{x - a}$$

$$\lim_{x \rightarrow a} \frac{\frac{1}{x^n} - \frac{1}{a^n}}{x - a}$$

$$\lim_{x \rightarrow a} \frac{\frac{a^n - x^n}{x^n a^n}}{x - a}$$

$$\lim_{x \rightarrow a} \frac{a^n - x^n}{x^n a^n} \times \frac{1}{x - a} \quad \lim_{x \rightarrow a} \frac{a^n - x^n}{x^n a^n} \times \frac{1}{x - a}$$

$$\lim_{x \rightarrow a} \frac{-(x^n - a^n)}{x^n a^n} \times \frac{1}{(x - a)} \quad \lim_{x \rightarrow a} \frac{a^n - x^n}{x - a} \times \frac{1}{x^n a^n}$$

$$= -n a^{n-1} \times \frac{1}{a^n}$$

$$= -n a^{n-1} \times \frac{1}{a^n}$$

$$= -n a^{n-1} \times \frac{1}{a^n}$$

ଉପରୋକ୍ତ ଉକ୍ତା ଉପାଦାନ ଉପରେ ଆଧାର କରି ଉକ୍ତ ଉପାଦାନ ଉପାଦାନ

$$(iii) \lim_{x \rightarrow 1} \frac{x'' - 1}{x - 1}$$

$$\lim_{x \rightarrow 1} \frac{x^{11} - 1^{11}}{x - 1}$$

$$(*) \lim_{x \rightarrow 1} \frac{x^2 + x^3 + x^{10} + \dots + x^{100}}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^2-1)(x^3-1) \cdots (x^n-1)}{x-1}$$

(G4) $\lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$

$$(5) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

$$(01) \lim_{x \rightarrow 1} \frac{x^{2016} - 1}{x^{2015} - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x^{2016} - 1}{x^{2015} - 1} \quad (02) \lim_{x \rightarrow 1} \frac{x^7 - 1}{x^6 - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x^{2016} - 1}{x^{2015} - 1} \quad (03) \lim_{x \rightarrow 2} \frac{x^4 - 16}{x^5 - 32}$$

$$= \lim_{x \rightarrow 1} \frac{2016 \times 1^{2015}}{2015 \times 1^{2014}} \quad (04) \lim_{x \rightarrow 1} \frac{3\sqrt{x} - 1}{\sqrt{x} - 1}$$

$$= \frac{2016}{2015} \quad (05) \lim_{x \rightarrow 1} \frac{x^5 - x^3}{x^3 + x^3}$$

$$(06) \lim_{x \rightarrow 1} \frac{4\sqrt{x} - 1}{3\sqrt{x} - 1}$$

$$(02) \lim_{x \rightarrow 1} \frac{x^7 - 1}{x^6 - 1}$$

$$\lim_{x \rightarrow 1} \frac{x^7 - 1}{x^6 - 1}$$

$$\frac{7 \times 1^{7-1}}{6 \times 1^{6-1}}$$

$$= \frac{7}{6}$$

$$(03) \lim_{x \rightarrow 2} \frac{x^4 - 16}{x^5 - 32}$$

$$\lim_{x \rightarrow 2} \frac{x^4 - 2^4}{x^5 - 2^5}$$

$$\lim_{x \rightarrow 2} \frac{x^4 - 2^4}{x^5 - 2^5}$$

$$= \frac{4 \times 2^4 - 1}{5 \times 2^5 - 1}$$

$$= \frac{4 \times 16}{5 \times 32}$$

$$= \frac{4 \times 8}{5 \times 32}$$

$$= \frac{1}{5}$$

$$(04) \lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}$$

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}$$

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{2} \times 1^{\frac{1}{2}-1}}{\frac{1}{2} \times 1^{\frac{1}{2}-1}}$$

$$\frac{1}{2} \times \frac{2}{1}$$

$$= \frac{2}{2}$$

$$(5) \lim_{x \rightarrow 1} \frac{x^5 - x^3}{x^2 - x^3}$$

$$\lim_{x \rightarrow 1} \frac{x^5 - x^3}{x^2 - x^3}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x^5} - \frac{1}{x^3}}{\frac{1}{x^2} - \frac{1}{x^3}}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1 - x^{10}}{x^5}}{\frac{1 - x^6}{x^3}}$$

$$= \lim_{x \rightarrow 1} \frac{1 - x^{10}}{x^5} \times \frac{x^3}{1 - x^6}$$

$$= \lim_{x \rightarrow 1} \frac{1 - x^{10}}{1 - x^6} \times \frac{1}{x^2}$$

$$= \lim_{x \rightarrow 1} \frac{1 - x^{10}}{1 - x^6} \times \frac{1}{x^2}$$

$$= \lim_{x \rightarrow 1} \frac{10 \times 1}{6 \times 1} \times \frac{1}{x^2}$$

$$= \frac{10}{6} \times \frac{1}{1}$$

$$= \frac{5}{3}$$

$$(6) \lim_{x \rightarrow 1} \frac{4\sqrt{x} - 1}{3\sqrt{x} - 1}$$

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}{x^{\frac{1}{3}} - 1^{\frac{1}{3}}}$$

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}{x^{\frac{1}{3}} - 1^{\frac{1}{3}}}$$

$$\frac{1}{2} \times \frac{\frac{1}{2} - 1}{\frac{1}{3} - 1}$$

$$\frac{1}{2} \times \frac{3}{1}$$

ଉତ୍ତର ଦିଅନ୍ତୁ ଏକ ଏକ ପ୍ରକାରର ଉତ୍ତର ଦିଅନ୍ତୁ.

$$(01) \lim_{x \rightarrow 64} \frac{3\sqrt{x} - 4}{\sqrt{x} - 8}$$

$$\lim_{x \rightarrow 64} \frac{x^{\frac{1}{3}} - 4 \times \frac{1}{3}}{x^{\frac{1}{3}} - (2^{3 \times \frac{1}{3}})^{\frac{1}{3}}}$$

$$\lim_{x \rightarrow 64} \frac{x^{\frac{1}{3}} - (64)^{\frac{1}{3}}}{x^{\frac{1}{3}} - (64)^{\frac{1}{3}}}$$

$$\lim_{x \rightarrow 64} \frac{x^{\frac{1}{3}} - 64^{\frac{1}{3}}}{x - 64}$$

$$\lim_{x \rightarrow 64} \frac{\frac{1}{3} \times 64^{\frac{1}{3} - 1}}{\frac{1}{3} \times 64^{\frac{1}{3} - 1}}$$

$$\frac{\frac{1}{3} \times 64^{\frac{1}{3} - \frac{2}{3}}}{\frac{1}{3} \times 64^{\frac{1}{3} - \frac{2}{3}}}$$

$$\frac{\frac{1}{3} \times 64^{-\frac{1}{3}}}{\frac{1}{3} \times 64^{-\frac{1}{3}}}$$

$$\frac{\frac{1}{3} \times 4^{-1}}{\frac{1}{3} \times 4^{-1}}$$

$$\frac{\frac{1}{3} \times 2^{-2}}{\frac{1}{3} \times 2^{-2}}$$

$$\frac{\frac{1}{3} \times 4^{-1}}{\frac{1}{3} \times 4^{-1}}$$

$$\frac{\frac{1}{3} \times 2^{-2}}{\frac{1}{3} \times 2^{-2}}$$

$$= \frac{\frac{1}{3} \times \frac{1}{16}}{\frac{1}{3} \times \frac{1}{8}} = \frac{\frac{1}{3} \times \frac{1}{16}}{\frac{1}{3} \times \frac{1}{8}} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

$$(02) \lim_{x \rightarrow 2} \frac{(x+a)^{2016} - (a+2)^{2016}}{x-2}$$

$$\lim_{x \rightarrow 2} \frac{(x+a)^{2016} - (a+2)^{2016}}{x+a - a-2}$$

$$\lim_{x \rightarrow 2} \frac{(x+a)^{2016} - (a+2)^{2016}}{x-2}$$

$$\frac{2016 \times (x+a)^{2015} \times (x+a - (a+2))}{2016 \times (x-2)^{2015}}$$

$$= \lim_{x+a \rightarrow a+2} \frac{(x+a)^{2016} - (a+2)^{2016}}{(x+a) - (a+2)}$$

$$= \frac{2016 (a+2)^{2015}}{1}$$

$$(03) \lim_{x \rightarrow 1} \frac{3\sqrt{x} + 3\sqrt{x} - 2}{x^5 - 1}$$

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0120
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$$\lim_{x \rightarrow 1} \frac{(x^2)^{\frac{1}{3}} + (x)^{\frac{1}{3}} - 2}{x^5 - 1}$$

$$\lim_{x \rightarrow 1} \frac{(x^2)^{\frac{1}{3}} + x^{\frac{1}{3}} - 2}{x - 1}$$

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{3}} - 1}{x - 1}$$

$$x^{\frac{1}{3}} = y \text{ put}$$

$$\lim_{x \rightarrow 1} \frac{(x^{\frac{1}{3}})^3 + x^{\frac{1}{3}} - 2}{x^5 - 1}$$

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{3}} - 1}{x - 1}$$

$$\lim_{x \rightarrow 1} \frac{y^3 + y - 2}{x^5 - 1}$$

$$\lim_{x \rightarrow 1} \frac{(x^{\frac{1}{3}} - 1)(x^{\frac{1}{3}} + 2)}{(x^5 - 1)}$$

$$\lim_{x \rightarrow 1} \frac{(x^{\frac{1}{3}} - 1)(x^{\frac{1}{3}} + 2)}{(x^5 - 1)}$$

$$\lim_{x \rightarrow 1} \frac{\left(\frac{1}{3} \times 1^{\frac{1}{3}-1}\right) \times 3}{5 \times 1^{5-1}}$$

$$\frac{\frac{1}{3} \times 1}{5} = \frac{1}{5}$$

I 400

$$(05) \lim_{x \rightarrow 0} \frac{(1+mx)^{\frac{1}{3}} - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{(1+mx)^{\frac{1}{3}} - 1^{\frac{1}{3}}}{x}$$

$$\lim_{x \rightarrow 0} \frac{[(1+mx)^{\frac{1}{3}} - 1^{\frac{1}{3}}]}{x}$$

$$= \lim_{x \rightarrow 0} \frac{[(1+mx)^{\frac{1}{3}}]^3 - [1^{\frac{1}{3}}]^3}{x [(1+mx)^{\frac{2}{3}} + (1+mx)^{\frac{1}{3}} + 1]}$$

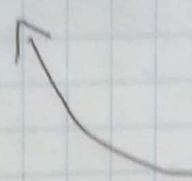
$$= \lim_{x \rightarrow 0} \frac{1+mx - 1}{x [(1+mx)^{\frac{2}{3}} + (1+mx)^{\frac{1}{3}} + 1]}$$

$$\lim_{x \rightarrow 0} \frac{m}{m/1+1+1} = \frac{m}{3}$$

$$x - y = (x^{\frac{1}{3}})^3 - (y^{\frac{1}{3}})^3$$

$$= (x^{\frac{1}{3}} - y^{\frac{1}{3}})$$

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} + x^{\frac{1}{3}}y^{\frac{1}{3}}$$



$$\frac{[(1+mx)^{\frac{2}{3}} + (1+mx)^{\frac{1}{3}} + 1^{\frac{2}{3}}]}{[(1+mx)^{\frac{2}{3}} + (1+mx)^{\frac{1}{3}} + 1^{\frac{2}{3}}]}$$

II 400

$$\lim_{x \rightarrow 0} \frac{(1+mx)^{\frac{1}{3}} - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{((1+mx)^{\frac{1}{3}} - 1^{\frac{1}{3}})m}{(1+mx) - 1}$$

$$\lim_{\substack{mx \rightarrow 0 \\ 1+mx \rightarrow 1 \\ 1+mx \rightarrow 1}} \frac{((1+mx)^{\frac{1}{3}} - 1^{\frac{1}{3}})m}{(1+mx) - 1}$$

$$\frac{1}{3} \times 1^{\frac{1}{3}-1} \times m$$

$$\frac{m}{3}$$

$x \rightarrow \alpha$ හි මිශ්‍රණය ක්‍රියාත්මක කිරීම බෙදීම

$x \rightarrow \alpha$ හි මිශ්‍රණය ක්‍රියාත්මක කිරීම බෙදීම කළහොත්
- පරිශ්‍ර ඉලක්කය වශයෙන් දෙකම 0, නිසාම ම.ප්‍ර.ප.වල බෙදීම
විෂයය.

$$(01) \lim_{x \rightarrow \alpha} \frac{x^2}{x^2 + 1}$$

$$\lim_{x \rightarrow \alpha} \frac{\frac{x^2}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}}$$

$$\lim_{x \rightarrow \alpha} \frac{1}{1 + \frac{1}{x^2}}$$

$$\frac{1}{1 + 0} = 1$$

$$(02) \lim_{x \rightarrow \alpha} \frac{2x^2}{1+x}$$

$$= \lim_{x \rightarrow \alpha} \frac{\frac{2x^2}{x^2}}{\frac{1}{x^2} + \frac{x}{x^2}}$$

$$\lim_{x \rightarrow \alpha} \frac{2}{\frac{1}{x^2} + 1}$$

$$\frac{2}{0 + 1} = 2$$

$$(03) \lim_{x \rightarrow \alpha} \frac{1+x+x^2}{1-x+x^2}$$

$$\lim_{x \rightarrow \alpha} \frac{\frac{1}{x^2} + \frac{x}{x^2} + \frac{x^2}{x^2}}{\frac{1}{x^2} - \frac{x}{x^2} + \frac{x^2}{x^2}}$$

$$\lim_{x \rightarrow \alpha} \frac{\frac{1}{x^2} + \frac{1}{x} + 1}{\frac{1}{x^2} - \frac{1}{x} + 1}$$

$$\frac{\frac{1}{\alpha^2} + \frac{1}{\alpha} + 1}{\frac{1}{\alpha^2} - \frac{1}{\alpha} + 1}$$

$$\frac{0 + 0 + 1}{0 - 0 + 1} = 1$$

$$(04) \lim_{n \rightarrow \infty} \frac{1+2+3+4+\dots+n}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{n}{2} (a+l)$$

$$S_n = \frac{n}{2} (a+l)$$

$$\lim_{h \rightarrow \infty} \frac{\frac{h}{2} (1+h)}{h^2}$$

$$\lim_{h \rightarrow \infty} \frac{\frac{h}{2} \times \frac{1}{h^2} (1+h)}{h^2}$$

$$\lim_{h \rightarrow \infty} \frac{1+h}{2h}$$

$$\lim_{h \rightarrow \infty} \frac{\frac{1}{h} + \frac{h}{h}}{2 \frac{h}{h}}$$

$$\lim_{h \rightarrow \infty} \frac{\frac{1}{h} + 1}{2}$$

$$\frac{1}{2}$$

$$(05) \lim_{x \rightarrow \alpha} \frac{x^3 + 1}{x^2 + 1}$$

$$= \lim_{x \rightarrow \alpha} \frac{(x+1)\sqrt{x^2}}{x^2 + \frac{1}{x^2}}$$

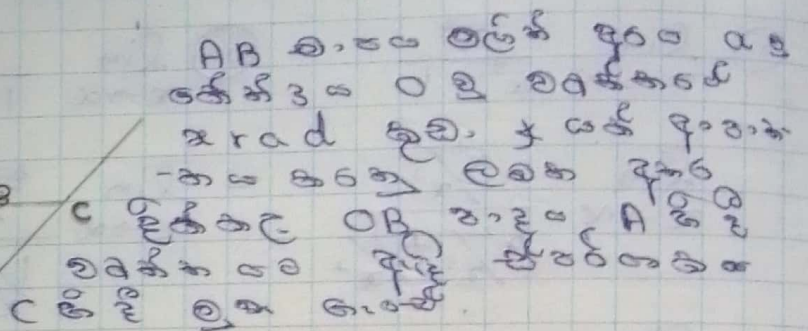
$$= \lim_{x \rightarrow \alpha} \frac{\frac{x}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \alpha} \frac{\frac{1}{x} + \frac{1}{x^2}}{1 + \frac{1}{x^2}}$$

$$= \frac{\frac{1}{\alpha} + \frac{1}{\alpha^2}}{1 + \frac{1}{\alpha^2}}$$

$$= \frac{0 + 0}{1 + 0} = \frac{0}{1} = 0$$

(କେଉଁ ଧରଣର ଫଳରାଶି ପାଇଁ)



$$\tan \alpha = \frac{AC}{a}$$

$$\frac{1}{2} r^2 \odot 2.64$$

$$\frac{1}{2} \times a \times a \sin \alpha < \frac{1}{2} \times a^2 \times \alpha < \frac{1}{2} \times a \times a \tan \alpha$$

$$\sin x < x < \tan x$$

$$\sin \alpha < x < \frac{\sin \alpha}{\cos \alpha}$$

$$\frac{\sin x}{\sin x} < \frac{x}{\sin x} < \frac{\sin x}{\cos x} \times \frac{1}{\cancel{\sin x}} \quad (\because \sin x > 0)$$

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}$$

$$1 > \frac{x}{\sin x} > \frac{1}{\cos x}$$

$1 < 2 < 3$
 $\frac{1}{1} > \frac{1}{2} > \frac{1}{3}$

$$\lim_{x \rightarrow 0} 1 > \lim_{x \rightarrow 0} \frac{\sin x}{x} > \lim_{x \rightarrow 0} \cos x$$

$$1 > \frac{\sin x}{x} > 1$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \text{Q.E.D.}$$

$$\textcircled{01} \lim_{x \rightarrow 0} \frac{\sin 3x}{x}$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right) \times \frac{1}{x} \times \frac{3x}{1}$$

$$\frac{1 \times 3}{3} \left\{ \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 1 \right\}$$

$$\textcircled{03} \lim_{x \rightarrow 0} \frac{\tan x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{\cos x} \times \frac{1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{1}{\cos x}$$

$$= \lim_{x \rightarrow 0} 1 \times \frac{1}{\cos x}$$

$$= \underline{\underline{1}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x - x}{\sin 3x + x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{x} - \frac{x}{x}}{\frac{\sin 3x}{x} + \frac{x}{x}}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{x} - 1}{\frac{\sin 3x}{x} + 1}$$

$$\textcircled{02} \lim_{x \rightarrow 0} \frac{\sin 3x - x}{\sin 3x + x}$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x - x}{\sin 3x + x} \left(\frac{\sin 3x - x}{\sin 3x - x} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right) \times \frac{3x - x}{\sin 3x + x}$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right) \times \frac{3x + x}{\sin 3x + x}$$

$$\lim_{x \rightarrow 0} \frac{1 \times 3x - \lim_{x \rightarrow 0} x}{\lim_{x \rightarrow 0} 1 \times 3x + \lim_{x \rightarrow 0} x}$$

$$\lim_{x \rightarrow 0} \frac{3x - x}{3x + x}$$

$$\frac{3x}{3x} = \underline{\underline{1}}$$

$$= \lim_{x \rightarrow 0} 1 - x$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{x} - 1$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} + 1$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{x} \times \frac{3x}{3x} \right) - 1$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{x} \times \frac{3x}{3x} \right) + 1$$

$$\frac{3-1}{1}$$

$$= \frac{2}{1}$$

$$= \underline{\underline{2}}$$

4) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2}$$

$$\lim_{x \rightarrow 0} 2x \left(\frac{\sin x}{x} \right)^2$$

$$2 \times 1$$

$$\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x}$$

$$\sin x = y \text{ proof,}$$

$$\lim_{y \rightarrow 0} \frac{\sin y}{y} = 1$$

5) $\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{x}$

$$= \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{x} \times \frac{\sin x}{\sin x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \times \frac{\sin(\sin x)}{\sin x}$$

$$= 1 \times 1$$

$$= 1$$

* 6) $\lim_{x \rightarrow 0} \frac{1 - \cos(2 \sin x)}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos(2 \sin x)}{x^2} \times \frac{2 \sin x}{2 \sin x}$$

$$2 \sin x = 0 \text{ proof,}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos(2 \sin x)}{2 \sin x} \times \frac{2 \sin x}{x^2} \times x$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin \left(\frac{2 \sin x}{2} \right)}{2 \sin x} \times \frac{2 \sin^2(\sin x)}{x^2}$$

$$= \lim_{x \rightarrow 0} 2 \left[\frac{\sin(\sin x)}{\sin x} \right] \times \left[\frac{\sin(\sin x)}{\sin x} \right] \times \frac{\sin^2 x}{x^2}$$

$$= 2 \times 1 \times 1$$

$$= 2$$

$$(07) \lim_{x \rightarrow 0} \frac{\tan 2x + \sin 2x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin 2x}{\cos 2x} + \frac{\sin 2x}{1}}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin 2x - \sin 2x \cos 2x}{\cos 2x}}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x (1 - \cos 2x)}{\cos 2x \cdot x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x \times 2 \sin^2 x}{\cos 2x \cdot x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x \times 2 \sin^2 x}{x^3 \times \cos 2x}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{\sin 2x}{2x}\right) \times \frac{2x}{1} \times \frac{1}{\cos 2x} \times \frac{2 \times \sin^2 x}{x^2} \times \frac{x^2}{1}}{x^3 \times \cos 2x}$$

$$= \lim_{x \rightarrow 0} \frac{1 \times 2x \times 2 \times 1 \times x^2}{x^3 \times \cos 2x}$$

$$= \lim_{x \rightarrow 0} \frac{2x^3 \times 2}{\cos 2x \times x^3}$$

$$= \lim_{x \rightarrow 0} \frac{4x^3}{x^3} \times \frac{1}{\cos 2x}$$

$$= 4 \times 1$$

$$= \underline{\underline{4}}$$

$$= \lim_{x \rightarrow 0} -2 \sin \frac{2x}{2} \times \frac{1}{2x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x}\right) \times \frac{3x}{2x} \times \frac{1}{x} = -\frac{3}{2}$$

$$(08) \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{17+x} - \sqrt{17}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{(\sqrt{17+x} - \sqrt{17}) \times (\sqrt{17+x} + \sqrt{17})}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x (\sqrt{17+x} + \sqrt{17})}{(\sqrt{17+x})^2 - (\sqrt{17})^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x (\sqrt{17+x} + \sqrt{17})}{17+x - 17}$$

$$= \frac{1 \times \sqrt{17+0} + \sqrt{17}}{1}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{17+x} + \sqrt{17}}{1}$$

$$= \underline{\underline{2\sqrt{17}}}$$

$$(09) \lim_{x \rightarrow 0} \frac{\cos 3x - \cos x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-2 \sin \frac{3x}{2} \times \sin \frac{x}{2}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-2 \sin \frac{3x}{2} \times \sin \frac{x}{2}}{\frac{x^2}{4} \times \frac{1}{1}}$$

$$= \lim_{x \rightarrow 0} \frac{-2 \sin \frac{3x}{2} \times \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}}\right) \times \frac{x}{2}}{\frac{x^2}{4} \times \frac{1}{1}}$$

$$= \lim_{x \rightarrow 0} \frac{-2 \sin \frac{3x}{2} \times 1}{\frac{3x}{2}} \times 1$$

$$(10) \lim_{x \rightarrow a} \frac{\sin(a-x)}{a-x}$$

$$= \lim_{x \rightarrow a} \frac{\sin(a-x)}{a-x}$$

$$= \lim_{y \rightarrow 0} \frac{\sin y}{y}$$

$$= 1$$

$$\therefore \lim_{x \rightarrow a} \frac{\sin(a-x)}{a-x} = 1$$

$$(11) \lim_{x \rightarrow 1} \frac{\sin(x^2+x-2)}{x^2-1}$$

$$= \lim_{x \rightarrow 1} \frac{\sin(x+2)(x-1)}{(x-1)(x+1)}$$

$$= \lim_{x \rightarrow 1} \frac{\sin(x+2)(x-1)(x+2)}{(x-1)(x+1)(x+2)}$$

$$= \lim_{x \rightarrow 1} \frac{\sin(x+2)(x-1)(x+2)}{(x-1)(x+2)(x+1)}$$

$$= 1 \times \frac{3}{2}$$

$$= \frac{3}{2}$$

$$(12) \lim_{x \rightarrow \pi} \frac{\sin x}{\pi^2 - x^2}$$

$$= \lim_{x \rightarrow \pi} \frac{\sin x}{(\pi-x)(\pi+x)}$$

$$= \lim_{x \rightarrow \pi} \frac{\sin x}{x} \times \frac{x}{(\pi-x)(\pi+x)}$$

$$= \lim_{x \rightarrow \pi} \frac{\sin(\pi-x)}{\pi-x} \times \frac{1}{(\pi+x)}$$

$$= 1 \times \frac{1}{2\pi}$$

$$= \frac{1}{2\pi}$$

$$(13) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1-\cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{x}{\sqrt{2\sin^2 \frac{x}{2}}}$$

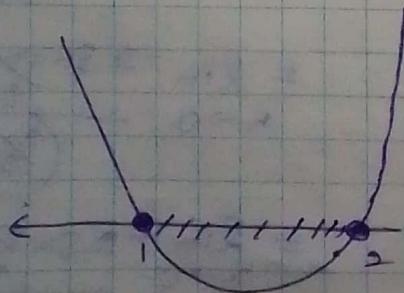
$$= \lim_{x \rightarrow 0} \frac{x}{\sqrt{2} \sin \frac{x}{2}}$$

$$= \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$= \sqrt{2}$$

(14) $x^2 - 3x + 2 \leq 0$ లో x కి గల విలువలను నిర్ణయించండి.

$$(x-2)(x-1) \leq 0$$



అందుకే x కి విలువలు 1 మరియు 2 మధ్య ఉంటాయి.

$$x \in [1, 2]$$

20) $\lim_{h \rightarrow 0} \frac{\sin(a+3h) - 3\sin(a+h) + 3\sin(a-h) - \sin a}{h^3}$

$$\lim_{h \rightarrow 0} \frac{\sin(a+3h) - \sin a - 3[\sin(a+h) - \sin(a-h)]}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{a+3h+a}{2}\right)\sin\left(\frac{a+3h-a}{2}\right) - 3\left[2\cos\frac{a+h+a}{2}\sin\frac{a+h-a}{2}\right]}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{2a+3h}{2}\right)\sin\left(\frac{3h}{2}\right) - 3\left[2\cos\frac{2a+h}{2}\sin\frac{h}{2}\right]}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{2a+3h}{2}\right)\sin\frac{3h}{2} - 6\cos\left(\frac{2a+h}{2}\right)\sin\frac{h}{2}}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{2a+3h}{2}\right) \times \left\{\sin\frac{3h}{2} - 3\sin\frac{h}{2}\right\}}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{2a+3h}{2}\right) \times \left[3\cancel{\sin\frac{h}{2}} - 3\sin\frac{h}{2} - 3\cancel{\sin\frac{h}{2}}\right]}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{2a+3h}{2}\right) \times -4\left(\frac{\sin\frac{h}{2}}{\frac{h^3}{8}}\right) \times \frac{1}{8}}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{2\cos\left(\frac{2a+3h}{2}\right) \times -32}{h^3} \cdot \lim_{h \rightarrow 0} \cos\left(\frac{2a+3h}{2}\right)$$

$$2 \times \cos\left(\frac{2a+3 \times 0}{2}\right) \times -32$$

$$2 \times \cos\left(\frac{2a}{2}\right) \times -32 = -64\cos a$$

Answer: $\underline{\underline{-64\cos a}}$

$$(98) \lim_{x \rightarrow 0} \frac{x}{(1+2x)^{\frac{1}{4}} - 1}$$

$$\lim_{x \rightarrow 0} \frac{x}{(1+2x)^{\frac{1}{4}} - 1}$$

$$\lim_{x \rightarrow 0} \frac{1}{\frac{(1+2x)^{\frac{1}{4}} - 1}{x}}$$

$$\lim_{x \rightarrow 0} \frac{x/2x}{(1+2x)^{\frac{1}{4}} - 1}$$

$$\lim_{x \rightarrow 0} \frac{1}{\left[\frac{(1+2x)^{\frac{1}{4}} - 1}{1+2x-1} \right] \times 2}$$

$$\lim_{x \rightarrow 0} \frac{1+2x = y}{x \rightarrow 0 \text{ so } y \rightarrow 1}$$

$$\lim_{y \rightarrow 1} \frac{1}{\left[\frac{(y)^{\frac{1}{4}} - 1}{y-1} \right] \times 2}$$

$$\lim_{y \rightarrow 1} \frac{1}{\left(\frac{y^{\frac{1}{4}} - 1}{y-1} \right) \times 2}$$

$$= \frac{1}{\frac{1}{4} \times 1^{\frac{1}{4}-1} \times 2}$$

$$= \underline{\underline{\frac{2}{1}}}$$

$$(18) \lim_{h \rightarrow \infty} \frac{1+2+3+\dots+h}{h^2}$$

$$S_n = \frac{n}{2} (a+l)$$

$$S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$\lim_{h \rightarrow \infty} \frac{\frac{h}{2} (1+h)}{h^2}$$

$$\lim_{h \rightarrow \infty} \frac{1}{2} \frac{(1+h)}{h}$$

$$\lim_{h \rightarrow \infty} \frac{1}{2} \left(\frac{1}{h} + 1 \right)$$

$$\frac{1}{2} [0+1]$$

$$\frac{1}{2} \lim_{h \rightarrow \infty} \frac{1}{h} + 1$$

$$\frac{1}{2} \left[\frac{1}{\infty} + 1 \right]$$

$$\underline{\underline{\frac{1}{2}}}$$

(24)

No

$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - \sqrt[3]{x} + 1}{(x-1)^2}$$

$$\lim_{x \rightarrow 1} \frac{(x^2)^{\frac{1}{3}} - 2 \times (x)^{\frac{1}{3}} + 1}{(x-1)^2}$$

$$\lim_{x \rightarrow 1} \frac{(x^{\frac{1}{3}})^2 - 2 \times (x^{\frac{1}{3}}) + 1}{(x-1)^2}$$

$$\lim_{x \rightarrow 1} \frac{y^2 - 2y + 1}{(x-1)^2}$$

$$\lim_{x \rightarrow 1} \frac{(y-1)^2}{(x-1)^2}$$

Date:

$$\frac{1}{x^3} = y \text{ } \neq 0$$

$$\lim_{x \rightarrow 1} \frac{(x^{\frac{1}{3}} - 1)^2}{(x-1)^2}$$

$$\lim_{x \rightarrow 1} \left[\frac{x^{\frac{1}{3}} - 1^{\frac{1}{3}}}{x-1} \right]^2$$

$$\lim_{x \rightarrow 1} \left[\frac{1}{3} \times 1^{\frac{1}{3}-1} \right]^2 = \frac{1}{9}$$

$$\underline{\underline{\frac{1}{9}}}$$